

LAB Assignment:- 02

Q1) What are the different protocols you observe at the following layers of the protocol stack?

- A. Application layer
- B. Transport layer
- C. Network layer

Ans $\Rightarrow$ 1) Application layer

- HTTP (used for http://neverssl.com)
- HTTP/TLS (used for https://www.cornell.edu)
- DNS (domain name resolution)

2) Transport layer

- TCP (used by HTTP and HTTPS)
- UDP (used by DNS)

3) Network layer

- IPv4

Q.2) What is the total amount of data being received for the following two cases?

- A. when you access http://neverssl.com/
- B. when you access http://www.cornell.edu

Ans $\Rightarrow$ A. $\rightarrow$ Total data received: 2512 bytes (≈ 2.45 KB)

B. $\rightarrow$ Total data received: 17,444 bytes (≈ 17.0 KB)

Q.3 Answer the following when you access the site <http://neverssl.com/>?

A) How many HTTP GET requests do you observe?
List down the GET requests.

Ans → Total HTTP GET requests: 3

- 1) GET /online HTTP/1.1
- 2) GET /online/ HTTP/1.1
- 3) GET /favicon.ico HTTP/1.1

B) What version of HTTP the server is running?

- HTTP version: HTTP/1.1
- Server: Apache/2.4.66

C) For each of the HTTP GET requests, find out:

- (i) The total number of TCP segments being received
- (ii) The total amount of data being received in the corresponding HTTP response message.

Ans →

HTTP GET Request	TCP Segments Received	Total Data Received
GET /online	3	2512 bytes
GET /online/	0 (redirect handled by previous response)	0 bytes
GET /favicon.ico	0	0 bytes

D) Identify TCP three way handshake used to establish the connection with the <http://neverssl.com>. Specify SYN, SYN-Ack, ACK packets and write the sequence number for each of the three access?

Ans → TCP Three-way Handshake

Packet	Frame No.	Sequence Number
SYN	15514	170255736
SYN-Ack	15329	2355175658
ACK	15330	170255737